

Space-bending interaction: mastering deformations of non-euclidean spaces to cheat at golf

Romain Chabbert¹Rémi Coulon²Lysa Gramoli¹Maé Mavromatis¹Florian Nouviale¹Adrien Reuzeau³Mathieu Risy¹Sabrina Toofany³Ronan Gaugne³Valérie Gouranton¹

¹Univ Rennes, INSA Rennes, Inria, CNRS, IRISA ²Université de Bourgogne, CNRS, IMB ³Univ Rennes, Inria, CNRS, IRISA

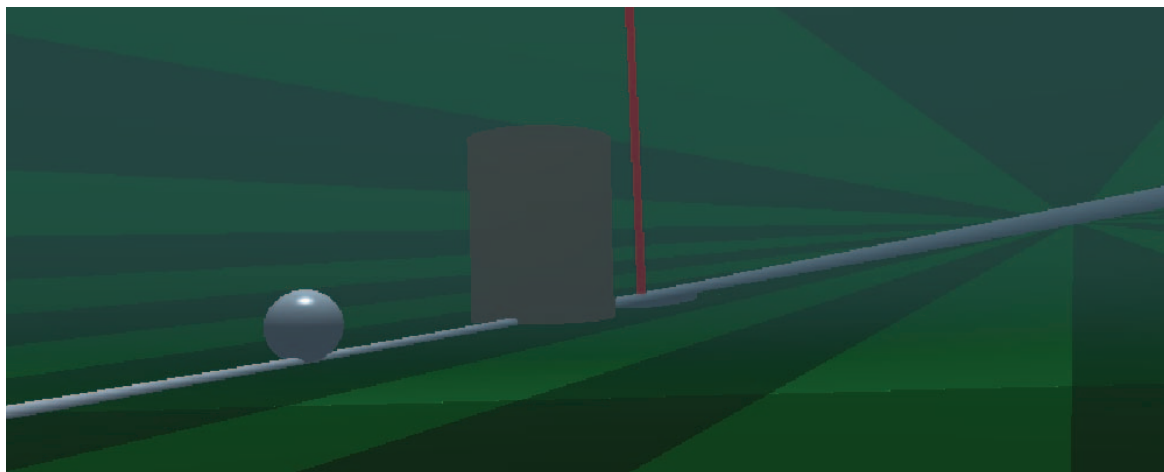


Figure 1: View of the golf course in spherical space. The white sphere is the golf ball, the brown cylinder is an obstacle in front of the hole marked by the red peg. The white line is the geodesic followed by the ball. The green stripes correspond to the grass covering the golf course. They also appear on the ceiling because of the properties of spherical space.

ABSTRACT

Some tasks that do not have solutions in euclidean space have solutions in other non-euclidean spaces. For this edition of the IEEEVR 3DUI contest, we are offering users the chance to bend euclidean space using a 3D piston slider in virtual reality (VR). This will make it possible to solve mini-golf problems that are impossible to solve in euclidean geometry, using spherical geometry.

Index Terms: Human-centered computing—Virtual reality; Applied computing—Interactive learning environments Mathematics of computing—Geometric topology

1 INTRODUCTION

We propose an original virtual experience to explain the deformation of the euclidean space into spherical space. This complex mathematical concept is illustrated by a mini-golf course whose success requires the deformation of euclidean space into spherical space. The innovative interaction of the system is based on a control of the curvature of the space through a push-pull piston. The deformation of the space is visualized in real time.

The spherical geometry is one example of “non-euclidean” geometries whose principles are presented in Section 2. This should not be confused with the (improper) use of the word “non-euclidean” to refer to portals or impossible spaces [3], often encountered in the game industry. Many programs simulate in virtual reality what it would be like to live in such a space [1, 2]. These works rely on the Fermat principle: light travels along geodesics. As these paths may not be straight lines, it severely affects our perception of the surrounding environment.

The complexity of these environments makes an intuitive understanding difficult to achieve. Few works have proposed mathematically consistent interactions in such environments to provide the user with a more intuitive knowledge of the properties of these geometries [6, 7]. To the best of our knowledge, none has illustrated the effect of interactive curvature modification in virtual reality.

2 UNDERLYING MATHEMATICAL CONCEPTS

Riemannian geometry is a field of mathematics that studies topological shapes together with a metric, i.e. a way to measure the length of any curve. In two dimensions, we are familiar with the euclidean plane, which corresponds to the geometry one usually learn in middle school. Other geometries are possible though, such as the 2-dimensional sphere, in which the path minimizing the distance between two points (called a *geodesic*) is no more a “straight” line. The geometrization theorem stated by Thurston and proved by Perelman provides a description of all possible geometries in 3 dimensions. Roughly speaking, it says that the understanding of 3-dimensional spaces boils down to the study of eight geometric models called *Thurston geometries*. Among them, one finds the 3D euclidean space, which we are familiar with but also seven other *non euclidean geometries*, such as the 3D sphere.

Curvature is an important concept in riemannian geometry. It is a scalar valued function κ which controls among other things how the space “bends” explaining why geodesics are not straight lines. For this software we decided to focus on the interplay between euclidean geometry and spherical geometry. In the euclidean space, the curvature is always zero. In a 3-dimensional sphere of radius r this function is constant equal to $\kappa = 1/r^2$. If we deform the space

by increasing the radius, the curvature tends to zero and we recover at the limit the euclidean space.

Remark: Because of our game scenario, we are not exactly working in the 3-dimensional sphere, but rather in an orbifold quotient modelled on the sphere. One can think of such a space as a “spherical pacman”. The actual space is a half sphere. As soon as an object reaches the boundary of this space, it is automatically teleported to a mirrored position. However this teleportation is just a technical machinery: for a user immersed in this space, the aforementioned boundary is not perceivable

3 DESIGN AND IMPLEMENTATION

We designed a virtual environment to visualize the non euclidean space, associated with an interactive game to understand and take advantage of the underlying mathematical properties.

3.1 Virtual environment

The project was developed using Unity3D and VR is handled through its OpenXR integration. The virtual environment is rendered using custom raymarching shaders where the rendering of basic shapes (half-space, sphere, cylinder, ...) is redefined for spherical space. We added fog to make the environment more legible, and to make explicit the parts of the environment that are furthest from the user.

3.2 Gameplay

The user has to make a single mini-golf shot, which aims to hit with the ball the hole indicated by a red peg. The scenes are designed in such a way that the target cannot be reached in euclidean space because of an obstacle in front of the hole, forcing the user to manipulate the curvature of space and take advantage of certain properties of spherical space to make a successful shot.

The user has access to two levels of difficulty corresponding to the number of obstacles to be avoided in order to make a successful shot.

4 INTERACTIONS

The application allows two interactions, one to control the curvature of space, and the other to make a golf shot. These interactions have been designed to be performed close to the user, in a range where space can be approximated to a euclidean space.

4.1 Space curvature control through a push-pull piston

The interaction used to control the curvature of space was created with a number of constraints in mind: control the parameter continuously so that the user can better understand the effects of their actions, keep the manipulations close to the user in a range where space can be approximated to a euclidean space, and the interaction must give the user the sensation of pushing/pulling space.

We created a push/pull piston-like (Figure 2) interaction mechanism that provides the user control over the curvature of the surrounding space. The piston is attached to the user left hand to be easily accessible. They can adapt their control by either moving the hand holding the mechanism, moving the hand interacting with it, or moving both at the same time. This piston-like mechanism provides the user with two visual feedback in addition to “bending” the space curvature. A linear gauge, similar to a 3D-slider metaphor [4], indicates the current value set by the user relative to the min (empty gauge) and max (full gauge) values. In addition, the user can grab the mechanism and deform it along one axis to either increase (push) or decrease (pull) the current value when going past a threshold (Figure 2). When the push-pull piston is released by the user, it shrinks/expands back to its original size.

We chose this interaction in the form of a piston to give the user the sensation of pulling or pushing the world away. The corresponding gesture, physically more involved than a simple slider, gives the user a feeling of superior control. This impression could be further



Figure 2: Push-pull piston interaction for space parameter control. Left : empty gauge. Right : gauge being filled while the mechanism is pulled

enhanced by modifying the ratio between the movement performed and the filling of the gauge, to turn the piston into a pseudo-haptic tool [5].

When they start the virtual reality experience, users are in euclidean space, i.e. in a space with zero curvature. As soon as they pull the piston, the curvature increases and the space is now spherical. The increase in the curvature of space corresponds to a decrease in the radius of the sphere on which the user is standing. When in spherical space, the user can also press the slider to perform the opposite operation and reduce the curvature of space until 0 is reached, in other words euclidean space.

4.2 Golf shooting interaction

There are two phases to the golf shot. First, the user chooses the direction in which they want to send the ball i.e. selects the geodesic that determines the ball’s trajectory. They can see the geodesic, and change its orientation using the controller’s joystick. Then they make a striking gesture with the putter. This gesture determines the speed of the ball, and triggers its departure.

5 CONCLUSIONS

We have proposed a virtual reality experience that allows users to interact in a non-euclidean space. The rules governing these environments are so diverse, and so different from euclidean space, that the proposed interactions are the result of strong arbitrations. In order to avoid unexpected behaviors, as described in [6], we chose to keep the interactions within the user’s neighborhood. Other choices could be explored, to allow the observation and understanding of these complex mathematical spaces. This same virtual experiment could be extended in other non-euclidean geometries, to solve other problems, in particular by exploiting the diversity of geodesic shapes in different geometries with negative curves, quotient, or heterogeneous dimensions.

ACKNOWLEDGMENTS

This work was supported in part by grants from CNRS 80 Prime ThurstonVR, DemoES AIR ANR-21-DMES-0001, Equipex+ Continuum ANR-21-ESRE-0030

REFERENCES

- [1] R. Coulon, E. A. Matsumoto, H. Segerman, and S. J. Trettel. Ray-Marching Thurston Geometries. *Experimental Mathematics*, 31(4):1197–1277, Oct. 2022.
- [2] R. Hudson, C. G. Gunn, G. K. Francis, D. Sandin, and T. A. DeFanti. Mathenautics: using vr to visit 3-d manifolds. In *ACM Symposium on Interactive 3D Graphics and Games*, 1995.
- [3] A. Murty and A. Glennerster. Route selection in non-euclidean virtual environments. *PLOS ONE*, 16(4):1–23, 04 2021.
- [4] D. Patterson. 3d spirals, bubbles and sliders: setting range values in multi-user 3d environments. In *Proc. of the Australasian Computer Science Week Multiconference*, pp. 1–8. ACM.
- [5] Y. Ujitoko and Y. Ban. Survey of pseudo-haptics: Haptic feedback design and application proposals. *IEEE Transactions on Haptics*, 14(4):699–711.
- [6] J. Weeks. Body coherence in curved-space virtual reality games. *Computers & Graphics*, 97:28–41, 2021.
- [7] C. Widdowson and R. F. Wang. Human navigation in curved spaces. *Cognition*, 218:104923.